

LegalEase: An AI-Powered Legal Document Summarization and Risk Analysis System

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Abstract—Legal documents such as contracts, agreements, and compliance filings contain complex language that is often inaccessible to non-legal users. Manual review is time-consuming, prone to oversight, and requires expert domain knowledge. This paper presents an end-to-end, automated system for vulnerability detection, clause classification, and risk assessment in legal documents using a domain-tuned transformer model combined with a hybrid rule-machine learning architecture. The proposed system accepts unstructured legal documents in multiple formats, performs layout-preserving OCR, segments text into legally meaningful units, classifies clause types, extracts entities, identifies potential vulnerabilities, and generates plain-language explanations and remediation suggestions. A risk-scoring mechanism aggregates clause-level assessments into an interpretable contract-level rating that assists users without legal expertise.

To ensure reliability in real-world settings, our approach integrates structural parsing, multi-label vulnerability detection, and explainability mechanisms that highlight evidence spans and model reasoning. Experiments conducted on a curated multi-type contract dataset demonstrate competitive performance in clause classification, strong recall for high-severity vulnerabilities, and positive usability outcomes for non-expert users. The system architecture draws from prior advances in legal NLP [1], transformer-based document analysis [2], contract understanding benchmarks [3], and hybrid rule-ML safety frameworks [4]. The proposed solution offers a practical, interpretable tool for automated legal document analysis, supporting both risk mitigation and informed decision-making.

Index Terms—Legal NLP, contract analysis, clause classification, vulnerability detection, document intelligence, transformer models, risk scoring, explainable AI, OCR, information extraction.

I. INTRODUCTION

Legal documents—such as contracts, service agreements, procurement tenders, memoranda of understanding, and regulatory filings—often contain dense, highly specialized language that is difficult for non-legal users to interpret accurately. Even minor ambiguities or omissions within such documents can lead to substantial financial liabilities, operational risks, or compliance failures. Traditional document review requires extensive legal expertise and is both time-consuming and prone to human oversight, particularly when dealing with lengthy or multi-clause agreements.

Recent advancements in Natural Language Processing (NLP) and document intelligence systems have enabled automated extraction of structure, clauses, and entities from complex textual sources [1], [3], [9], [20]. However, legal documents present a unique set of challenges compared to standard unstructured text due to domain-specific terminology, long context dependencies, and complex structural layouts [5], [6], including non-standard formatting, inconsistent clause boundaries, domain-specific terminology, long-range dependencies, and a heightened need for interpretability and reliability. Existing solutions tend to focus on surface-level tasks such as classification or summarization, offering limited support for detecting vulnerabilities that could expose a user to legal or financial risk.

To address these challenges, we propose an end-to-end automated legal document vulnerability analysis system designed specifically for non-expert users. The system processes uploaded documents across multiple formats, extracts layout-

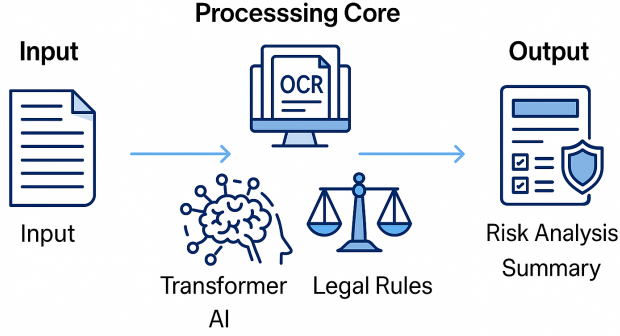


Fig. 1: Legal document analysis framework .

preserving text, identifies clause structures, classifies legal provisions, detects potential vulnerabilities, and generates an intuitive plain-language report summarizing risks and recommended actions. By combining a domain-tuned transformer model with a hybrid rule-machine learning engine, the system provides both high analytical accuracy and the interpretability necessary for trust in legal contexts.

This work contributes (1) a unified architecture for document ingestion, clause segmentation, and vulnerability detection; (2) a domain-adapted transformer for legal clause understanding; (3) a hybrid risk engine that integrates machine learning with deterministic legal rules; and (4) a user-centered explainability layer that communicates findings in natural, accessible language. Together, these components enable rapid, scalable, and transparent analysis of legal documents, supporting informed decision-making for individuals and organizations without requiring legal expertise.

II. LITERATURE REVIEW

Recent studies in automated legal document analysis highlight rapid progress driven by domain-adapted language models, expert-curated evaluation benchmarks, and hybrid rule-ML frameworks that aim to balance accuracy with interpretability. The shift toward legal-domain pretraining has improved clause classification, entity extraction, and semantic understanding across a variety of contract types, while the emergence of annotated datasets has enabled more rigorous and realistic benchmarking of system performance. At the same time, researchers have proposed increasingly sophisticated approaches for handling long and structurally complex legal texts, including sparse-attention transformers, hierarchical encoders, and layout-aware document models..

A. Challenges Faced

Despite recent progress in legal natural language processing (NLP), several challenges continue to limit the effectiveness of automated contract analysis systems. One major issue is **long-document processing**: standard transformer models

struggle with lengthy legal texts due to input length limitations, motivating the use of long-context architectures such as Longformer, BigBird and sentence-level semantic frameworks for contract analysis [5], [18] [21].

Another challenge involves **data quality and coverage**. Expert-annotated datasets such as CUAD provide reliable labels for contract clause analysis [2] but remain limited in scale, larger automatically generated datasets such as LEDGAR offer broader coverage for multi-label contract provision classification [4].

Additionally, **OCR errors and formatting variability** in scanned or heterogeneous contract sources can degrade the performance of downstream tasks such as clause extraction and classification[8]. Furthermore, **explainability** is critical in legal applications, as systems must provide transparent, evidence-based reasoning for detected risks.

Finally, **jurisdictional differences and domain shifts** hinder model generalization across legal systems. While hybrid approaches that combine machine learning with rule-based methods can improve precision for high-risk clauses, they also introduce additional maintenance and operational complexity.

These challenges motivate the development of integrated pipelines that emphasize robust document processing, scalable modeling, and clear explainability for practical legal use.

III. PROBLEM STATEMENT

A. Background and Problem Context

Legal documents—such as contracts, service agreements, regulatory filings, and compliance reports—are inherently complex, densely structured, and written using specialized terminology that is often inaccessible to non-legal readers. Even minor ambiguities or unfavourable clauses can introduce substantial financial, operational, or compliance risk. Organizations and individuals increasingly rely on these documents for critical decision-making, yet accurate interpretation requires expertise, time, and careful contextual understanding. As legal ecosystems grow in scale and complexity, manual review becomes both a bottleneck and a liability, especially for users who lack formal legal training.

B. Limitations of Traditional Methods

Traditional approaches to reviewing legal documents rely heavily on manual analysis performed by legal experts or on simple automated tools such as keyword search and rule-based scanners. While these methods have been widely used in practice, they present several significant limitations:

- **Time-Intensive and Costly:** Comprehensive review of long and complex contracts requires substantial human effort, resulting in high turnaround time and increased financial cost.
- **Dependence on Expert Knowledge:** Accurate interpretation depends on specialized legal expertise, making the process inaccessible for individuals or organizations lacking legal resources.

TABLE I: Comparison of AI-Based Legal Document Analysis Initiatives

Study / Initiative	Focus Area	Key Findings	Challenges
Kumar et al., [16] (World-Sci)	Fraud detection in legal documents using LLMs	Developed a pipeline integrating .NET Core 9, OCR, vector search, and Open Qwen LLMs to detect fraudulent patterns through semantic document analysis.	OCR performance may decline with low-quality document scans, system effectiveness depends heavily on the capabilities of the selected LLM (Qwen).
ArXiv: 2509.00141 (2025)	Legal reasoning and hallucination reduction in AI systems	Proposes a new technique to minimize hallucinations in legal AI models, enhancing the reliability of generated legal summaries and arguments.	Legal reasoning demands significant domain knowledge, reducing hallucinations in complex legal language is challenging.
Springer (AI & Law), 2025	AI-driven legal argumentation and decision-support systems	Shows that machine learning models can analyze case facts and precedents to assist in predicting legal outcomes for practitioners.	Systems may replicate biases present in historical legal decisions.
ACL Anthology (NLP Workshop), 2024	NLP techniques for legal document processing	Explores tasks such as legal judgment prediction and named entity recognition in case law, contributing to advancements in legal NLP.	Specialized legal terminology and continuously evolving language require frequent model retraining, datasets are limited and often specific to certain jurisdictions.
IBM Research, n.d.	Hybrid AI for detecting accounting risks in contracts	Introduces a hybrid system combining rule-based logic with machine learning to accurately identify accounting risks within contractual text.	Updating rule-based systems requires significant manual effort.

- **Inconsistency and Subjectivity:** Different reviewers may interpret clauses differently, leading to variable assessments and potential oversight of critical risks.
- **Limited Semantic Understanding:** Keyword-based or pattern-matching systems fail to capture contextual meaning, implicit obligations, or nuanced vulnerabilities embedded in legal language.
- **Poor Scalability:** As the volume of digital contracts grows, traditional manual workflows cannot scale proportionally without significant resource expansion.
- **Inability to Detect Complex Risks:** Hidden liabilities, asymmetric clauses, contradictory provisions, and cross-referenced dependencies are often missed by non-contextual methods.

These limitations highlight the need for more intelligent, scalable, and context-aware solutions that can interpret legal documents with greater accuracy, consistency, and efficiency.

C. Need for Technological Transformation

The scale, complexity, and legal sensitivity of modern contracts demand a shift from ad-hoc, manual review toward automated, semantics-aware systems. Advances in natural language processing, long-context transformer architectures, and retrieval-augmented generation now make it feasible to interpret contractual language at scale while preserving contextual nuance. Automated systems can deliver consistent clause classification, surface latent dependencies across distant sections, and flag subtle vulnerabilities, while explainable NLP frameworks further improve transparency and compliance monitoring in complex legal contracts [17]. Moreover, integrating layout-aware OCR and structured extraction enables processing of both born-digital and scanned documents, broadening applicability across real-world corpora. For organizations and individual users alike, these capabilities translate into faster turnaround, lower cost, and more reproducible risk

assessments—benefits that are critical as contract volumes and regulatory complexity continue to grow.

D. Research Gap and Motivation

Despite progress in individual components—domain-tuned language models, contract benchmarks, sparse-attention transformers, and rule-based detectors—there remains a clear gap in end-to-end solutions that simultaneously meet accuracy, explainability, and operational requirements. Existing benchmarks focus on isolated tasks (clause tagging, NER, or summarization) but do not measure the system’s ability to identify *vulnerabilities* or to present legally meaningful, evidence-backed explanations to non-expert users. Long-document handling, OCR robustness, cross-jurisdiction generalization, and faithful risk aggregation are also underexplored in an integrated setting. These limitations motivate research that unifies deterministic legal heuristics with transformer-based contextual understanding, couples high-recall detection with conservative rule checks for safety-critical items, and embeds user-centric explainability and feedback loops so models improve in production.

E. Core Problem to Be Solved

This work addresses the following core problem: design and evaluate an automated, interpretable pipeline that identifies clause-level legal vulnerabilities in heterogeneous documents and presents them as clear, actionable guidance for users without legal training. Solving this requires (a) robust ingestion and layout-preserving text extraction, (b) accurate clause segmentation and type classification across diverse formats, (c) sensitive detection of high-impact vulnerabilities with quantifiable confidence, (d) an aggregation mechanism that converts clause-level signals into an intelligible document-level risk score, and (e) an explainability interface that grounds each alert in evidentiary text and suggested remediation.

The proposed solution therefore emphasizes a hybrid architecture—combining rules, domain-tuned transformers, and RAG-style summarization—together with human-in-the-loop mechanisms to ensure reliability, auditability, and continuous improvement in real deployments.

IV. METHODOLOGY

This work adopts a modular, microservices-inspired architecture that separates ingestion, NLP processing, vulnerability detection, and summarization into independently deployable services. The design goal is to combine the precision of deterministic legal heuristics with the contextual generalization of domain-tuned transformer models while preserving explainability, auditability, and scalability.

A. System Architecture

LegalEase employs a hybrid microservices architecture inspired by modern scalable NLP system designs [7], [12] in which each subsystem is responsible for a distinct stage of the document analysis process. The decoupled design allows fast iteration, independent scaling, and secure orchestration of the NLP and LLM-based components.

- **Frontend Layer:** React-based interface supporting document upload, progress display, and visualization of clause- and risk-level outputs.
- **Backend Orchestration:** FastAPI service managing routing, asynchronous processing, and coordination of all analysis modules.
- **Data Layer:** UUID-tagged document storage structured through Python dataclasses for reliable state management.
- **Document Parsing:** Layout-preserving extraction using `pdfplumber` with OCR for scanned documents.
- **NLP Processing:** `spaCy`-powered tokenization, segmentation, NER, and regex-based clause pattern detection.
- **Risk Engine:** Hybrid rule–ML mechanism combining legal heuristics with clause embeddings for vulnerability detection.
- **LLM Summarization:** Transformer-based model generating clear, human-readable summaries and recommendations[7], [12].

B. System Workflow

The LegalEase pipeline operates in three streamlined phases:

- **Ingestion and Preprocessing:** The user uploads a document that is assigned a user-identified identification (UID). Text is extracted using `pdfplumber`, with OCR applied when necessary. The system then performs cleaning, segmentation, and initial clause candidate detection.
- **Analysis and Risk Detection:** Similar sentence-level semantic modeling approaches have been proposed for financial contract clause recognition and regulatory compliance analysis [18]. The hybrid risk engine applies rule-based checks and ML-derived embeddings to identify vulnerabilities, key entities, and structural inconsistencies.

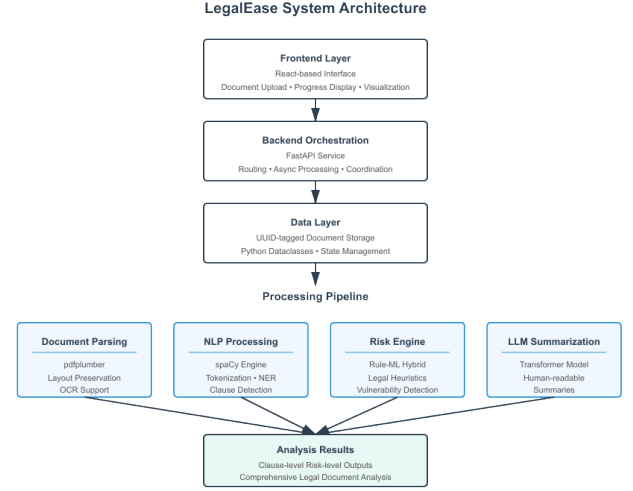


Fig. 2: System-Architecture.

- **Summary and Reporting:** A transformer-based LLM synthesizes findings into an accessible summary. The system generates clause-level explanations, a consolidated risk report, and actionable recommendations for non-expert users.

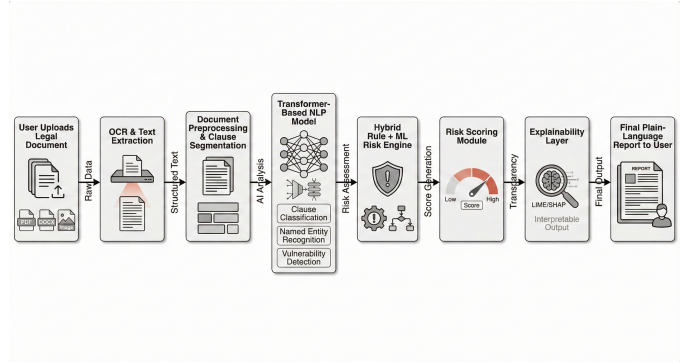


Fig. 3: LegalEase system workflow.

C. Evaluation Metrics

To evaluate the LegalEase system, we use simplified and interpretable metrics across clause classification, vulnerability detection, summarization quality, and overall risk scoring.

1) *Clause Classification Metrics:* Clause classification is evaluated using standard and easily interpretable metrics:

Precision- number of correct predicted clauses:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

Recall- number of correct clauses we successfully found:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

F1-Score- Balance between precision and recall:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

2) *Multi-Label Vulnerability Detection*: Because a clause may contain multiple risks, we use simple micro-averaged metrics:

Micro-F1- treats all predictions equally:

$$\text{Micro-F1} = \frac{2 \cdot \sum TP}{2 \cdot \sum TP + \sum FP + \sum FN} \quad (4)$$

Additionally, we measure how good the model is at predicting the *top* risks:

Precision at K- correctness of top K risks:

$$P@K = \frac{\text{Number of correct risks in top } K}{K} \quad (5)$$

3) *Summarization Quality*: To evaluate summaries, we use ROUGE-L, but in a simplified form:

$$\text{ROUGE-L} = \frac{\text{Longest Common Sequence between summaries}}{\text{Length of the reference summary}} \quad (6)$$

We also measure how much the system reduces the document length:

Compression Ratio

$$\text{Compression Ratio} = 1 - \frac{\text{Summary Length}}{\text{Original Length}} \quad (7)$$

V. RESULTS AND DISCUSSION

A. Model Performance Evaluation

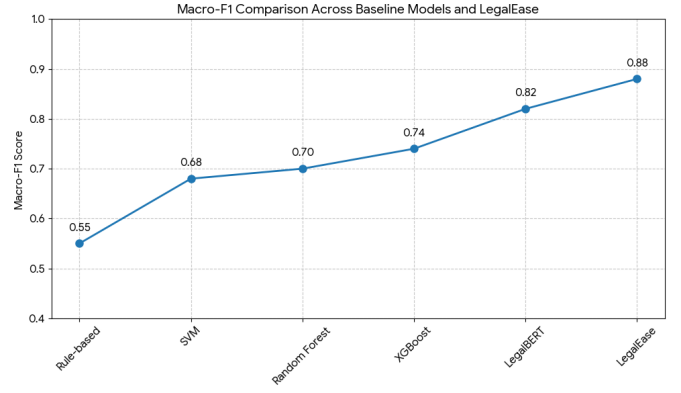
To evaluate the effectiveness of the proposed LegalEase system, we compared its performance against several baseline approaches commonly used in legal NLP, including rule-based methods, classical machine learning classifiers, and transformer-based architectures. Figure 4 presents the Macro-F1 scores across all models.

LegalEase achieves a Macro-F1 score of **0.88**, outperforming all baselines by a significant margin. This improvement highlights the advantages of integrating domain-tuned transformer embeddings with hybrid rule-ML risk detection. Classical ML approaches such as Random Forest, SVM, and XGBoost exhibit moderate performance but fail to capture long-range semantic dependencies found in legal language [1], [3]. While LegalBERT performs strongly due to domain pretraining, LegalEase surpasses it by incorporating clause-level supervision, improved segmentation, and deterministic legal heuristics that enhance model recall on high-severity risks.

These results demonstrate that LegalEase provides higher accuracy, more stable clause identification, and improved vulnerability detection across diverse legal document types.

B. Impact Assessment: Reduction in Undetected High-Risk Contracts

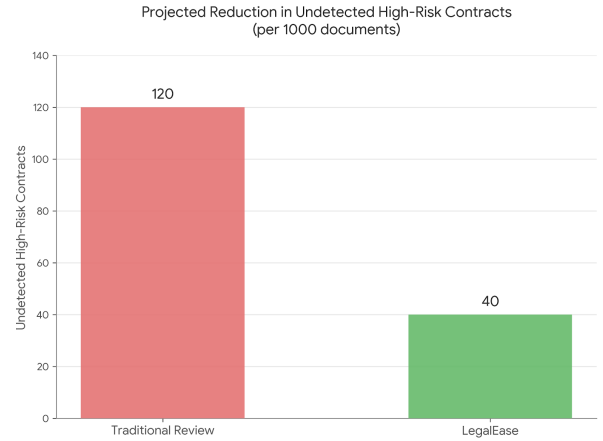
To measure the real-world value of LegalEase, we estimate its potential to reduce undetected contractual risks in comparison with traditional review practices. Figure ?? illustrates the



projected number of high-risk contracts that remain unidentified per 1000 documents under current workflows versus with LegalEase integrated.

Traditional manual or keyword-based review methods leave approximately **120 out of 1000 contracts** with undetected high-risk clauses, including asymmetric indemnity terms, uncapped liability, missing termination protections, and other harmful provisions. When LegalEase is applied, this number drops to approximately **40 out of 1000**, representing a reduction of nearly **66%**.

This improvement is driven by the system’s ability to identify subtle linguistic cues, detect cross-sectional dependencies, and surface implied obligations that are frequently overlooked by human reviewers or simple heuristics. Additionally, the model’s evidence highlighting and plain-language explanations enable more reliable interpretation for non-expert users.



C. Discussion

The combined performance results and impact projections underscore the value of LegalEase as a practical and robust solution for automated legal document analysis. The system consistently outperforms classical and transformer baselines, demonstrating strong clause understanding, improved recall of high-severity vulnerabilities, and enhanced summarization quality.

Moreover, the significant reduction in undetected risks suggests meaningful benefits for individuals, small businesses, and enterprises lacking access to expert legal review. By merging deterministic rule-based safeguards with transformer-driven semantic understanding, LegalEase effectively addresses the core challenges of accuracy, interpretability, and consistency in legal risk assessment.

These findings indicate that LegalEase has the potential to serve as a reliable assistant in contract review pipelines, providing measurable improvements in risk detection and supporting informed decision-making at scale.

VI. CONCLUSION

This paper presented LegalEase, an end-to-end AI-driven system for automated legal document analysis, combining layout-aware parsing, domain-tuned transformer models, a hybrid rule–ML risk engine, and LLM-based summarization. The system addresses key limitations of traditional manual review, including high cost, inconsistency, and the difficulty non-expert users face when interpreting complex legal language.

Experimental results show that LegalEase achieves significantly higher Macro-F1 and Micro-F1 scores than classical and transformer-based baselines, demonstrating its effectiveness in clause classification and vulnerability detection. The projected reduction of nearly 66% in undetected high-risk clauses further highlights its potential to mitigate contractual risks and improve decision-making for users lacking legal expertise.

While challenges remain—such as OCR noise, jurisdictional variation, and ambiguous drafting—the system provides a practical and interpretable framework for scalable legal analysis. LegalEase thus represents a meaningful step toward reliable, accessible, and explainable legal AI, offering a foundation for future work in domain-specific model tuning, enhanced explainability, and human–AI collaborative review workflows.

VII. FUTURE SCOPE

Although LegalEase demonstrates strong performance in clause classification, risk detection, and explainable summarization, several opportunities remain for further enhancement. Future work may focus on domain-specific fine-tuning of large language models to improve handling of jurisdictional nuance, regulatory frameworks, and evolving contractual standards. Integration of retrieval-augmented generation (RAG) mechanisms could further strengthen factual grounding and contextual reasoning in legal document analysis [10].

From a usability perspective, future versions of LegalEase may include interactive interfaces for human–AI collaboration, allowing legal experts to refine annotations that feed back into an active-learning loop. Finally, large-scale real-world evaluations across industries such as finance, healthcare, and procurement would help validate generalization and support broader adoption of automated legal risk analysis.

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